

14.3 Course Wrap

Question, tidy, plot, model, communicate. Iterate.

1. The lifecycle

A data analysis is a loop, not a straight line:

1. a question
2. tidy the table
3. plot
4. model
5. communicate

You go around more than once. A plot changes the question. A model sends you back to the table. The last step is a document someone else can rerun.

2. Tools you have now

Job	What you reach for
Rows, columns, groups	dplyr verbs, <code>across()</code> , <code>case_when()</code>
Pictures	ggplot2
Files in and out	readr
Wider / longer tables	tidyr
Two tables	joins
Strings, factors, dates	stringr, forcats, lubridate
A mean difference or a line	<code>t.test()</code> , <code>lm()</code>
A writeup	Quarto / R Markdown

Iteration — loops and `purrr::map` — is the last verb: do the same thing to many pieces.

3. Last lesson

Iterate. Save the file. Render again. The competence is not one clever plot. It is being able to change a step and get a new document that still runs.

You already have the verbs. Week 14 is the habit of applying them to many pieces at once.

4. Practice

Write five bullets of work you can do now that you could not do in week 1. Be specific (a verb and a data set), not "I learned R."